

Enhancing Readability with a Target-Aware Zooming Technique for Touch Surfaces

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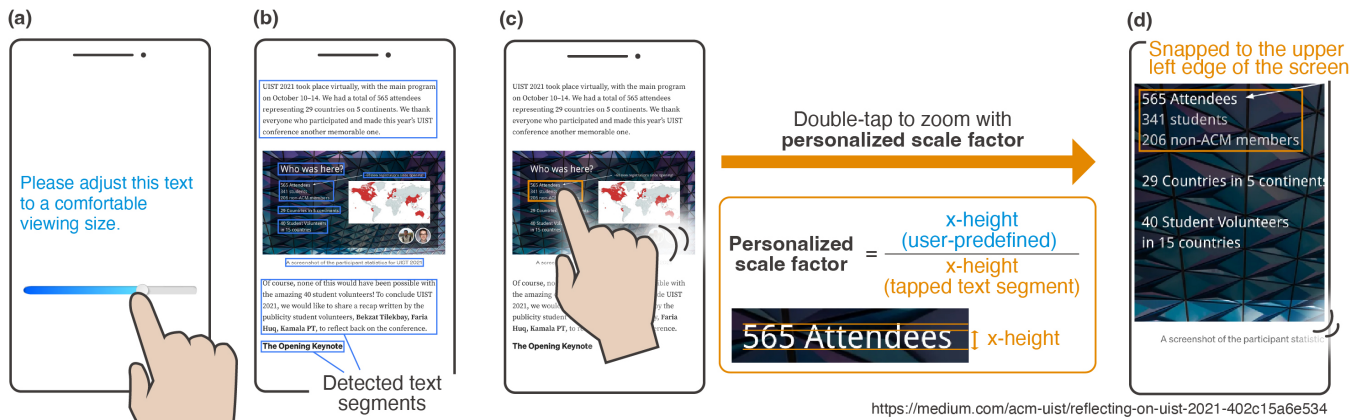


Figure 1: Overview of the proposed technique. (a) Users determine the readable text size in advance. (b) Our image viewer detects text segments within the content by using OCR. (c) When the user double-taps the text, the zoom level is automatically adjusted to the optimal level for reading the tapped text. The personalized scale factor is determined based on the user-defined text size (x-height) from (a) and the x-height of the text within the tapped segment. (d) Simultaneously, the top-left corner of the tapped segment snaps to the top-left corner of the screen, reducing the need to pan to adjust the position of the content.

ABSTRACT

Double-tapping is a common way to zoom in and out of content on touch surfaces, snapping between overview and detail levels. However, the optimal zoom level often differs for each user depending on the displayed content and user characteristics (e.g., age or visual acuity). When focusing on content containing text, the scaling factor is especially crucial for readability. However, the conventional double-tap to zoom uses a fixed scale factor, which often requires users to manually adjust the zoom level by pinching in and out after double-tapping. Additionally, on small-screen devices such as smartphones, the specific area of interest may not fit within the screen after zooming in, causing users to pan repeatedly to adjust the display position of the content. To address these issues, we propose a target-aware zooming technique that dynamically adjusts the zoom level based on the content and user preferences. Furthermore, to minimize the need for panning, our technique simultaneously snaps the top-left corner of the bounding box of the tapped text to the top-left corner of the screen as the content is zoomed in. This approach aims to reduce the need for manual

adjustments, improving usability and readability of digital content for diverse user groups.

CCS CONCEPTS

• Human-centered computing → Interaction techniques.

KEYWORDS

Double-tap to zoom, Readability, Text accessibility.

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1 INTRODUCTION

Touch interactions have become ubiquitous in modern devices. Despite their widespread adoption, there has been relatively limited focus on optimizing the canonical gesture set on smartphones or tablet PCs [1, 2]. Double-tapping to zoom is one of these fundamental touch gestures. It enables users to zoom in and out the displayed content, namely snapping between the overview and detail levels. In general, the optimal zoom level for users varies depending on the displayed content (e.g., font size and style) and user characteristics (e.g., age, visual acuity, and presbyopia) [3, 5, 7]. When focusing on content containing text, the scaling factor is especially crucial for

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readability; however, the conventional double-tap to zoom uses a fixed scaling factor. As a result, even after double-tapping, users often have to manually adjust the scale factor by pinching in and out. In addition, content such as digital comic books, catalogs, magazines, and flyers often have multiple sections of text with varying text sizes on the same page, further increasing the effort required to adjust the zoom level for comfortable reading. The iOS Books app¹ is one of the few examples that detects text in images and adjusts the scale factor accordingly. However, this enlarges the text to fit within the screen width instead of changing the scale factor to consider individual readability. Another aspect is that on small screen devices such as smartphones, the specific area of interest may not fit within the screen after zooming in, causing users to repeatedly pan to adjust the display position of the content [1].

We propose a target-aware zooming technique to avoid frequent pinch and pan operations. Since double-tapping to zoom has the advantages of being familiar, well-understood, and easy to perform even in single-handed situations, we further optimized this. Our system first retrieves a list of text sizes within the content. When a user double-taps on the content, the system dynamically adjusts the scale factor to match a pre-user-defined readable text size, taking into account the size of the text at the location where the touch-up event is detected. In addition, our technique snaps the top-left corner of the text's bounding box to the top-left corner of the screen simultaneously with zooming, users can immediately see the beginning of the text. Our technique is simple yet effective in optimizing the zoom level and position of the displayed content quickly without tedious manual adjustments and improving the readability of content for diverse user groups.

2 IMPLEMENTATION

We implemented a prototype of an image viewer application incorporating the proposed technique, running on iOS. In our application, the user's preferred text size is first set with a slider (Fig. 1 (a)). Next, the user selects images and opens it. The application utilizes Optical Character Recognition (OCR) to identify the positions of text segments within the page (Fig. 1 (b)), as it is designed to work with digital content such as comics, magazines, flyers, and catalogs in different formats (e.g., jpg, png, pdf). For each detected segment, the application records the height of characters included in the segment to determine the personalized scale factor. We used the x-height of characters at the zoom level set here (in px) as a baseline to adjust the scale factor to handle different font types (Fig. 1 (c)). This is because, even with the same font size, the width and height of characters differ for each font type, making the font normalization method based on x-height effective [10]. The application records the position and scale factor for each text segment in a list. When the user double-taps on a text segment, the scale factor is used to adjust the zoom level to match a user-defined readable size. In addition, by snapping the top-left corner of the text's bounding box to the top-left corner of the screen simultaneously with zooming², users can immediately see the beginning of the text, reducing the need for panning (Fig. 1 (d)). Note that our technique does not

change the layout of the content, only the zoom level and position of the entire content. To return to the original zoom level, the user can double-tap either the previously double-tapped segment again or any area without text.

3 ENVISIONED APPLICATIONS

In addition to taking user preferences into account, recognizing the context in which the user is using their smartphone can further enhance the zoom level.

User Posture-Aware Adjustments. Our technique, which dynamically adjusts the scale factor according to the user's preferences rather than fixing it, can be extended to use cases involving dynamic changes in the scale factor according to the user's situation. It is known that the face-to-screen distance varies significantly depending on the user's situation when using a smartphone [8]. For example, this distance is shorter when lying down, and longer when walking or sitting. Therefore, adjusting the personalized scale factor based on the user's situation will be promising. Specifically, we envision using methods that estimate the user's situation using the smartphone's internal sensors [6]. For example, based on the estimated situation, the predetermined text size is used when the user is sitting (Fig. 2 (a)), and the personalized scale factor is adjusted to a smaller value when the user is lying down (Fig. 2 (b)).

Environmental and Screen Brightness-Aware Adjustments. Screen brightness and ambient lighting can affect text visibility. Generally, lower brightness and darker surroundings require larger text for better visibility [4]. Similar to adjustments based on user activity, we envision further adjustments to the personalized scale factor based on screen brightness and ambient light sensor values. For example, in bright environments, the text size would match the predetermined size (Fig. 2 (c)), while in darker environments, a slightly larger scale factor would be used (Fig. 2 (d)).

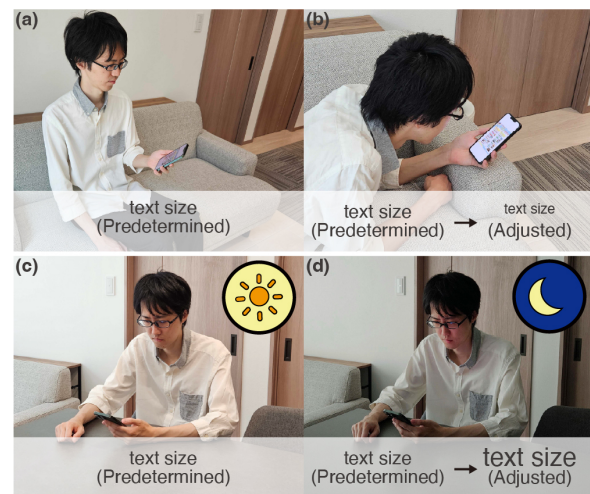


Figure 2: Envisioned applications. Our technique can be extended to use cases involving dynamic changes in the scale factor based on user activities and situations.

¹<https://apps.apple.com/us/app/apple-books/id364709193> (accessed August 5, 2024)

²In the initial settings, the snap position is set to the top-left corner. However, as individual preferences for reading areas on the screen can vary [9], we provided an option for users to change the snap position to their preferred location.

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